Review Article

Variation in Physicochemical Content in Rhizospheric Soil Samples of *Alstonia* Belongs to the Same Locations

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ABSTRACT

Alstonia scholaris (L.) R. Br. is an ornamental plant, commonly observed in gardens, also known as devil's tree or dita tree. The plant is considered poisonous, and birds rarely nest on it. It belongs to the family *Apocynaceae* and is native to Asia and Australia. Morphologically, this tall plant has funnel shaped flowers and glossy leaves. It holds cultural significance and is used in various medicinal applications. Rhizospheric soil samples were analyzed for several parameters including pH and EC measurement, estimation of nutrients, and chlorophyll content. The species *Alstonia scholaris* is present on the campus of SRTM University, Nanded, which predominantly has black clay soil that is suitable. These trees are located throughout the campus and are found in the Nanded city. The plants on campus belong to the same genus and species but display differences in color of leaves and physicochemical parameters of the rhizospheric soil. All the plants are growing close to each other approximately 10 meters. With the physicochemical analysis of the soil, there were differences in the nutritional composition of the rhizospheric soil. Among all selected parameters, pH, EC, carbon content, phosphorus content, potassium content, and chlorophyll were recorded as 7.3 to 8.5, 0.16 to 0.35 dS/m, 2.8 to 6.3×10^{3} mg/kg, 0.47 to 6.23 mg/kg, 34.9 to 927 mg/kg, and 9.72 to 21.97×10^{-3} mg/mL. Even when plants are in proximity, soil parameters differ. Plants do not receive the same nutrients. Variation in the pigmentation is a result of nutrient uptake by the plants reflected in leaf color variations.

Introduction

The *Alstonia scholaris* is planted throughout Nanded city, including SRTM University campus. These trees belong to the family *Apocynaceae* and can be used in the treatment of malaria, jaundice, gastrointestinal troubles, cancer, etc. [1].

Plants have played a vital role in maintaining human health and improving the quality of human life for thousands of years, providing

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valuable components for medicines, seasonings, beverages, cosmetics, and dyes. Herbal medicine contains natural substances that can promote health and reduce illness [2]. *A. scholaris* (L.) R.Br. is herbal medicine useful for numerous biological activities [3]. Potential bioactive compounds are present in all parts of the plant, including the stem bark, stem, and leaves. The plant and its parts are primarily used in the treatment of various illness [4,5], arthritis [6], infertility [7], wound healing and ear pain [8], respiratory disorders [9], women's health issues [10], dog bites [11], stomach problems [12,13], dental issues [14]. The active components of *Alstonia* are also used in the treatment of cancer, leprosy, mental disorders, cardiopathy, helminthiasis, pruritus, and agalactia [15]. The plant also falls under the category of aphrodisiac agents along with other well-known dry fruits [7]. Additionally, the plant is used to prepare antidotes to poison [16], and as an abortifacient [17]. The application of *Alstonia* also documented in veterinary practice for the treatment of cattle fever [18,19], phosphaturia and as a blood purifier [20].

Alstonia is commonly observed in Indian subcontinent, including Bangladesh, Sri Lanka, Nepal, Bhutan, and Pakistan, as well as parts of Southeast Asia, Australia and Queensland. In India, it can be found in Maharashtra, Karnataka, Kerala, Tamil Nadu, Bihar, Goa, etc. [1,21-23]. *A. scholaris* in urban areas can grow up to 20 m and spread around 10 while in natural habitat, it can reach heights of 50-60 m [24]. The bark and bark infusion are used in traditional medicinal practices as a cardiotonic and treatment of malaria and helminthiasis [25].

Materials and Methods

Seventeen plants were selected for the present investigations and rhizospheric soil was collected for the physicochemical analysis. The locations of these plants were geo-tagged and leaf samples were taken for the analysis as needed. Chlorophyll estimation was conducted using the methods of Talling and Driver [26], with the modifications by Mackinney [27].

Estimation of Nutrients and Mineral Contents

Around 100 gm soil samples from rhizosphere of each plant were taken and collected in sterile sampling polythene bags and stored in a portable cooling container untill they reached the laboratory. The samples were collected by digging the surface soil approximately 10 cm deep. The collected soil samples were air-dried and unnecessary components such as stones, dried grass, etc., were removed; the soil was ground with blender and passed through 2 µm sieves. The obtained soil was measured, kept it in cooling conditions [28]. This soil was further tested for the determination of pH, EC, carbon content, phosphorus, and potassium content. The pH and EC values were measured on-site using a portable Hanna pH and EC detection instrument.

Carbon

The carbon content present in the soil was determined by chromic acid wet oxidation method, with the amount measured in mg/kg [29].

Phosphorus

Phosphorus content present in the soil sample was detected using a colorimetric method. In this method, molybdenum reacts with phosphorus to form a blue colored compound measured at 882 nm. The concentration of phosphorus is directly proportional to the formation of the blue colored complex.

Potassium

The potassium content present in soil sample was determined by standard flame photometric method [28].

Results and Discussion

Physicochemical analysis of soil was conducted using standard methods and techniques, and the obtained results were evaluated through biostatistical analysis.

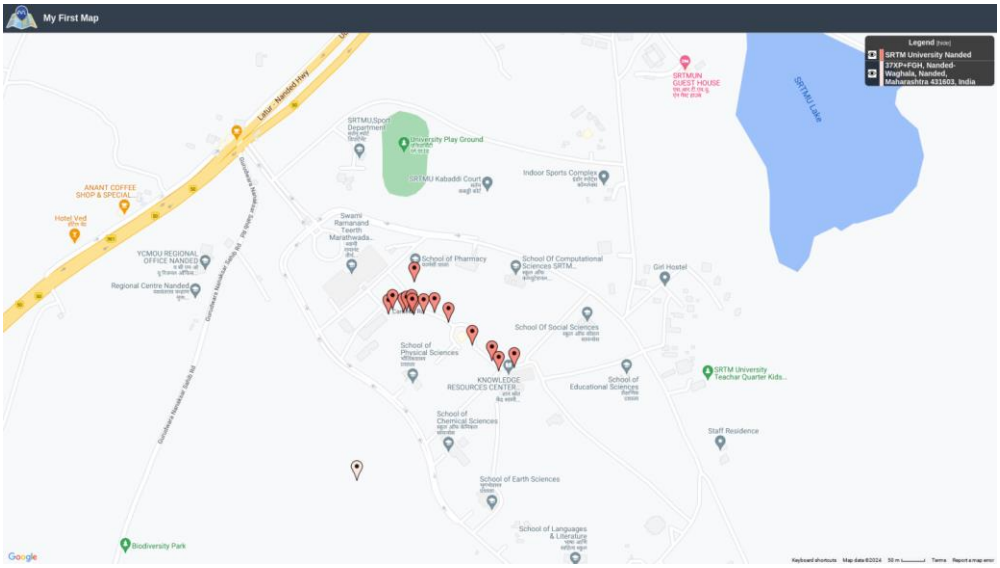


Figure 1: Sample location of SRTM university campus (Courtesy: The authors' first Map)

All the selected samples were located approximately 10 meters apart in the SRTM university campus (Figure 1, Table 1).

Table 1: Summary table of the raw values for all 17 samples

Sr./ No.	pH (± 0.01)	EC (dS/m)	Carbon Content (x10^3 mg/kg)	Phosphorus Content (x 10^{-1} mg/kg)	Potassium Content (x 10^{-1} mg/kg)	Chlorophyll Content (x10^{-3} mg/ml)
1	7.8	0.17	2.8	62.3	680	10.981
2	8.5	0.22	6.3	4.72	927	10.2303
3	8.3	0.32	3.2	40.77	470	9.7251
4	8.2	0.16	4.9	51.63	376	12.1248
5	8.3	0.16	4.2	51.63	632	14.9034
6	7.6	0.35	5.6	33.53	726	14.7771
7	8.2	0.26	5.9	22.98	470	15.7875
8	8.1	0.17	2.8	19.89	605	14.6508
9	7.8	0.19	5.6	15.58	403	9.7251
10	8.1	0.17	4.9	12.59	349	14.9034
11	8.0	0.21	5.6	15.11	470	12.63
12	7.3	0.32	4.9	16.53	349	13.6404
13	7.6	0.17	6.3	15.74	699	21.9762
14	8.2	0.18	3.5	17.32	618	12.5037
15	8.2	0.16	3.2	22.83	780	12.3774
16	8.1	0.18	4.9	10.7	927	12.3774
17	8.0	0.18	4.2	15.74	874	12.3774

pH Content

All seventeen plant samples were observed to have variation in pH scale in their rhizospheric soil. The samples show a pH range of the rhizospheric soil acidic to alkaline. The range of pH variations was observed to be 7.3 to 8.5 with standard error ±0.01 (Figure 2, Table 1). This pH range indicates that present soil is mostly slightly alkaline to strongly alkaline soil. The average soil pH variations were observed to be 6.0-7.5, which indicates the availability of optimum contents in the soil [30]. The pH is the main factor responsible for controlling active metabolism occuring within the soil. Based on pH variations, soil can be classified as ultra-acidic (<3.5), extremely acidic (3.5–4.4), very strongly acidic (4.5–5.0), strongly acidic (5.1–5.5), moderately acidic (5.6–6.0), slightly acidic (6.1–6.5), neutral (6.6–7.3), slightly alkaline (7.4–7.8), moderately alkaline (7.9–8.4), strongly alkaline (8.5–9.0), and very strongly alkaline (>9.0) [31,32]. In this regard, some of the soil samples fall within the range of moderately alkaline to strongly alkaline, while most soil samples fall into the neutral pH range.

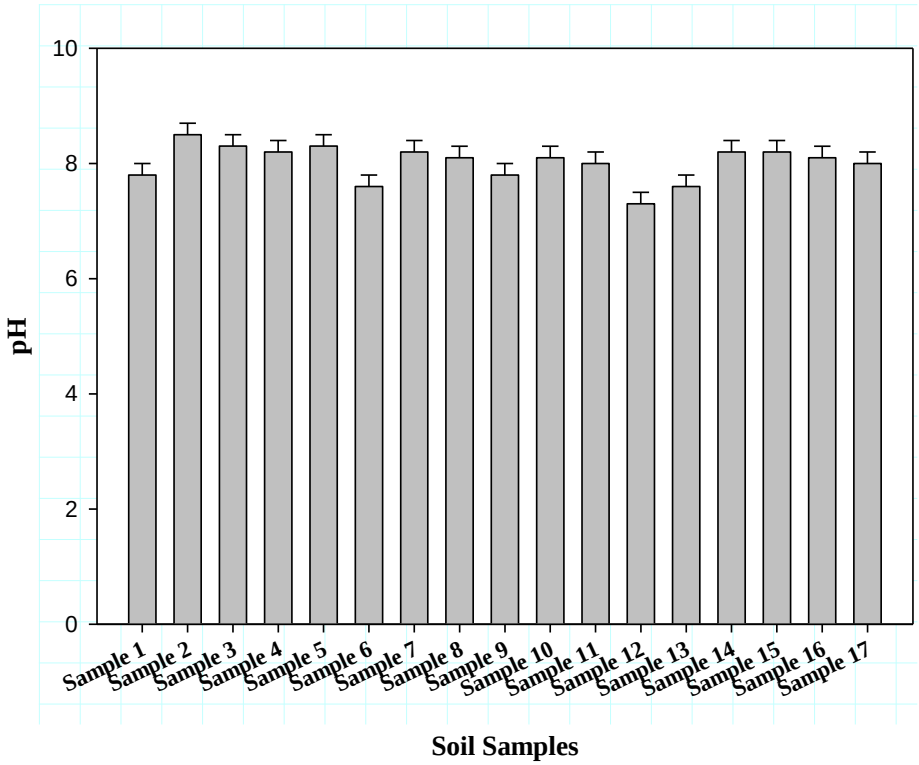


Figure 2: The pH variations in the collected soil samples of SRTM university campus

EC Content

In present investigation, all 17 soil samples were tested for the EC content, with the minimum EC content in samples 4, 5, and 15, which was around 0.16 ± 0.01 dS/m. The maximum EC content was observed in sample 6, which was around 0.35 ± 0.03 dS/m. Samples 3 and 12 also show an EC content around 0.32 ± 0.02 dS/m. The range of EC content in all 17 was 0.16 ± 0.01 dS/m to 0.35 ± 0.03 dS/m (Figure 3, Table 1). Soil conductivity is an important parameter to detect conduction or attenuation of electrical current; this parameter is mainly used for

detecting soil salinity. The presence of cations such as Ca^{2+}, Mg^{2+}, K^{+}, Na^{+}, and NH_4^{+} and anions such as SO_4^{2-}, Cl^{-}, NO_3^{-}, and HCO_3^{-} affects electrical conductivity of soil. Other factors responsible for the variations in the EC include soil organic matter, soil texture, temperature, and water content, etc. [33].

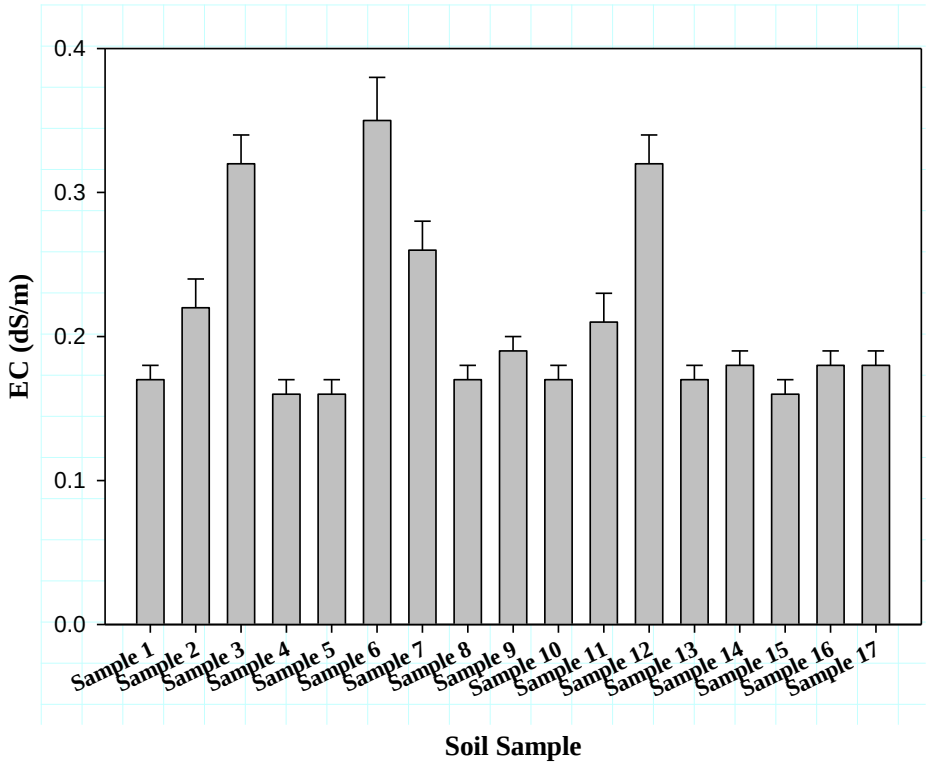


Figure 3: EC variations in the collected soil samples of SRTM university campus

Carbon content of Soil

The minimum carbon content detected in samples 1 and 8 was around 2.8 ± 0.2 x 10^3 mg/kg, while the maximum carbon content was detected in samples 2 and 13, which was around 6.3 ± 0.6 x 10^3 mg/kg of soil. The carbon content in all 17 soil samples were ranges from 2.8 ± 0.2 x 10^3 mg/kg to 6.3 ± 0.6 x 10^3 mg/kg of soil (Figure 4, Table 1). The major amount of organic matter present in the soil samples contains carbon and carbon derivatives. The organic matter which can be further categorized into dissolved, particulate, humus, and resistant [34].

Phosphorus Content

Phosphorus is an essential element for plants, along with nitrogen and potassium for their proper development and nutrition. The maximum phosphorus content was detected in sample 1, while the minimum content was observed in sample 2, which was around 6.23 ± 0.06 mg/kg and 0.47 ± 0.04 mg/kg, respectively. Numerous samples showed almost similar concentrations of phosphorus such as samples 8, 9, 10, 11, 12, 13, 14, and 16, which showed the P content as 1.9890 ± 0.010, 1.5580 ± 0.010, 1.2590 ± 0.010, 1.5110 ± 0.010, 1.6530 ± 0.010, 1.5740 ± 0.010, 1.7320 ± 0.010, 1.0700 ± 0.010, and 1.5740 ± 0.010 mg/kg, respectively. Samples 4 and 5 showed almost similar P content, which was 5.1630 ± 0.05 and 5.5630 ± 0.05 mg/kg (Figure 5, Table 1). The presence of phosphorus in soil can range from 500 to 800 mg/kg dry soil and is usually present in the form of orthophosphate. It can be associated with organic matter and mineral soils [35]. The average surface soil contains approximately 0.6% of P, while in case of sandy soils, it is recorded to be around 0.04%.

The major reasons for the variation in the nutrient contents of the rhizospheric soils are the densities and architecture of the roots, rate of nutrient uptake, and secretions of root exudates. The secretion was done by the roots most often in form of organic acids, amino acids, and sugars that affect the mobilization or immobilization of the nutrients. The probable other reasons for such variations are accumulation of leaf litter, differences in moisture retention, soil texture (compaction and aeration), and micro-topographical differences. The microbial flora such as phosphate solubilisation microbes, helps to release the nutrients like P from the soils. The variation in the pH directly affects the P content in the soil.

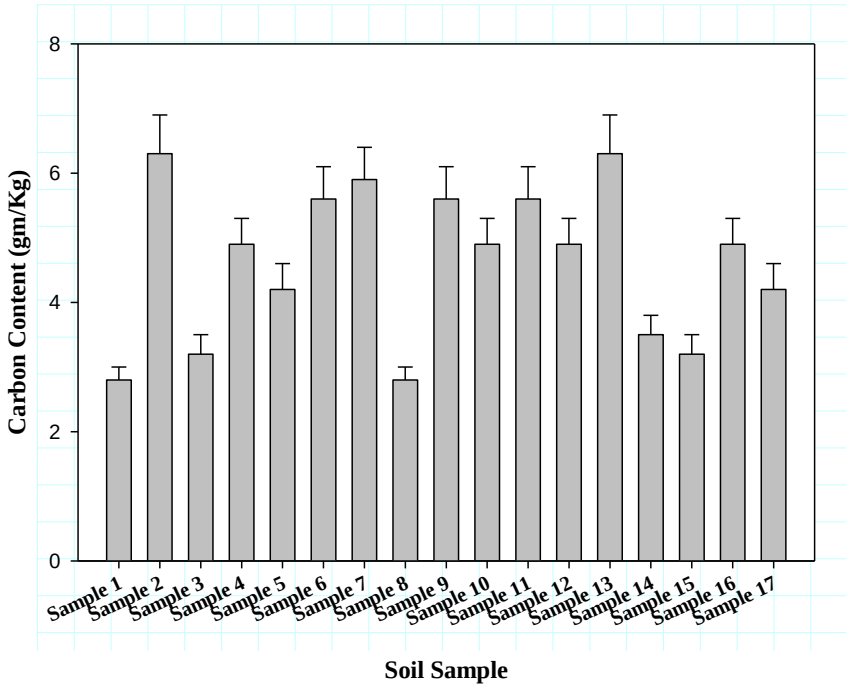


Figure 4: Carbon content in the collected soil samples of SRTM university campus

Potassium Content

Potassium is an important element with respect to plant growth and development. The maximum content of K was observed in samples 2 and 16, which was 927.00 ± 9.00 mg/Kg, while the minimum content was observed in sample 10, which was around 34.9 ± 3.0 mg/Kg. Samples 4 and 12 also showed approximately similar content, which were around 37.6 ± 3.0 mg/Kg and 34.9 ± 3.0 mg/Kg (Figure 6, Table 1). K is the major component involved in the development of roots and production of seeds. The insufficient provision of K leads to the lower yields and poor growth of roots, making crops more prone to disease infections and pest attacks [36-38]. The average amount of K present in Indian soil is around 0.5 to 3.00%. Plants are unable to utilize total available K present in the soil for their growth. The release of a high amount of K in the rhizosphere indicates presence of potassium-solubilizing bacteria (KSB). The presence KSB facilitates the release of K readily available to plants through weathering of complex minerals. The microbial secretions of organic acids lower the pH and promote weathering, acidolysis, and chelation of minerals [39]. The presence of KSB influences the percentage of K content in the soil. The microbial colonization of such microbes can create a drastic difference of 34.9 to 927 mg/kg within a short distance. The rhizosphere of black clay holds more nutrients and minerals.

Chlorophyll estimation

The maximum chlorophyll content was found 21.9762 x 10^{-3} mg/mL and the average chlorophyll content was approximately 13.2859 ± 2.9165 x 10^{-3} mg/mL (Figure 7, Table 1). The major component involved in the

photosynthetic activity of the plant is chlorophyll; thus, it is directly related to the growth of plants [40]. This investigation used the parameter of chlorophyll content as a variable with respect to variations in the nutritional components of the soil. The variation in the coloration of the leaves indicates that the supplements required for the growth of the plant can vary in the same locations of the rhizosphere.

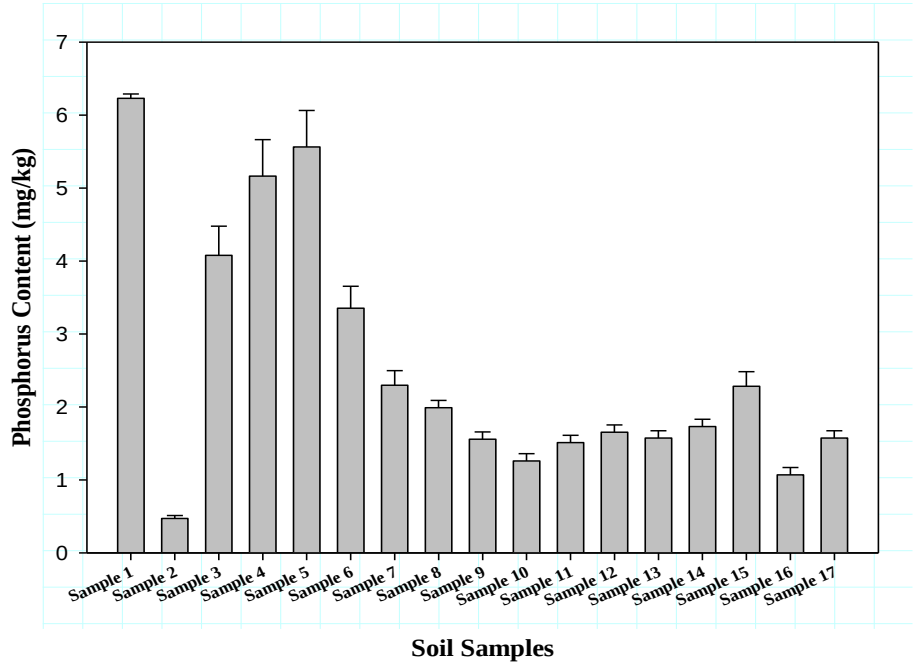


Figure 5: Phosphorus content in the collected soil samples of SRTM university campus

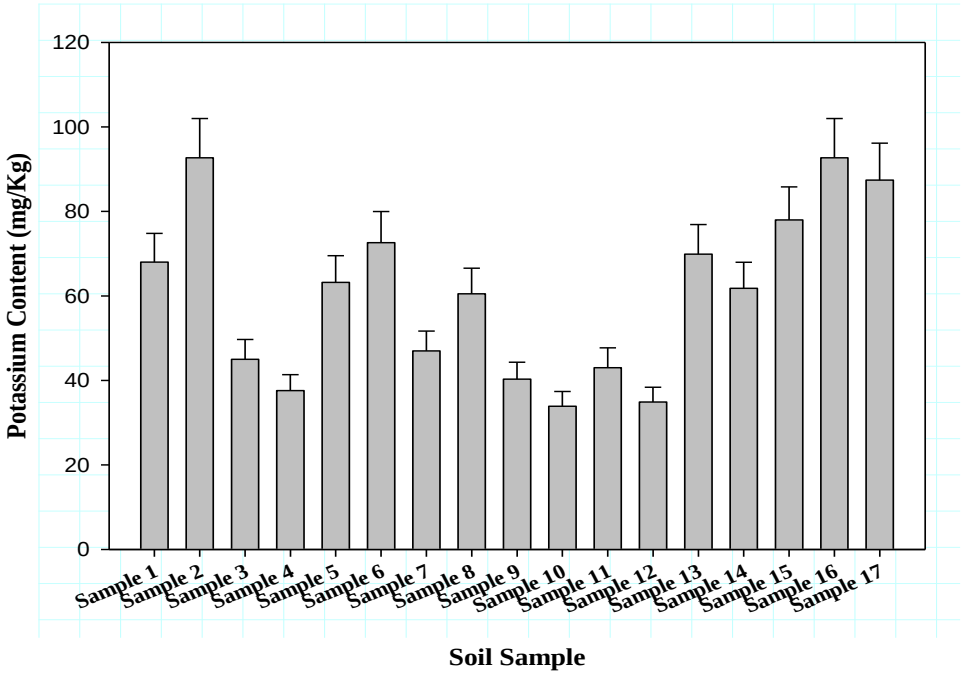


Figure 6: Potassium content in the collected soil samples of SRTM university campus

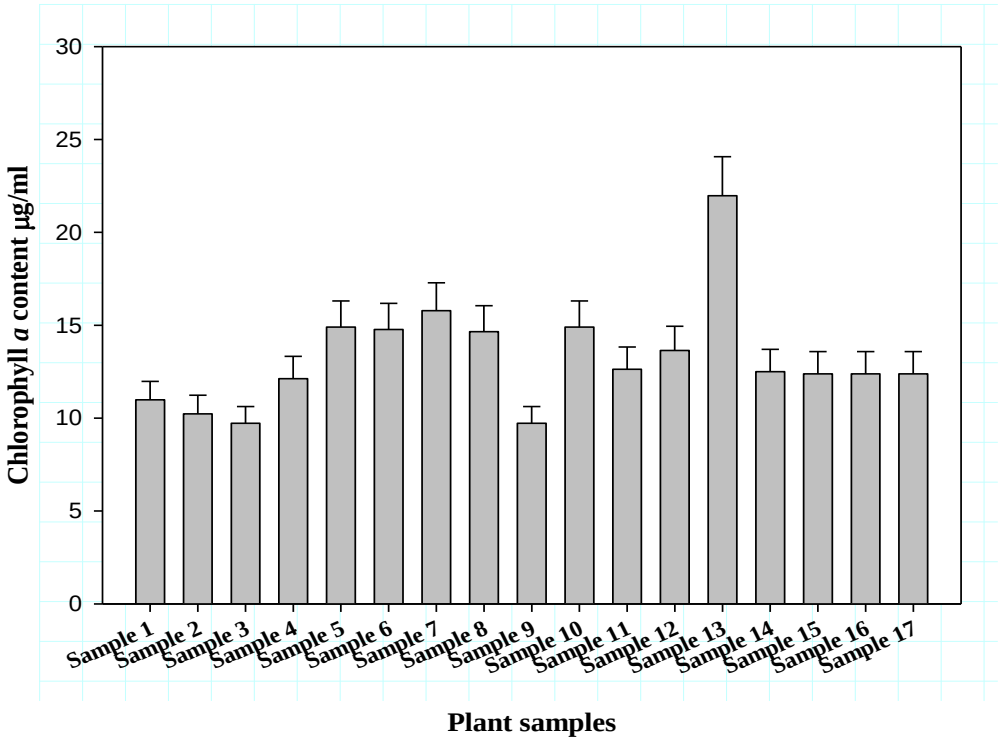


Figure 7: Chlorophyll content in the collected soil samples of SRTM university campus

Conclusion

In the current investigation, rhizospheric soil was collected and analyzed. Although sampling sites were close, they exhibited variations in their nutritional components. The same plant species (*Alstonia)* were planted on the campus of SRTM university, Nanded, but the plants did not equally absorb the elements of the soil.

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